

EDA Project – Suggested Analysis

Problem Scenario: Consider you are Music Director/Mixing Engineer aiming to optimize new songs for popularity, leveraging insights from this dataset is key. Here are some questions categorized by analysis type:

1. Univariate analysis :

- What is the overall distribution of popularity scores across all tracks in the dataset? (Are most songs moderately popular, or is it skewed towards very high/low popularity?)
- What is the average and typical range for duration_ms (song length)?
- What are the most frequently occurring keys in the dataset, and what is their individual distribution?
- How are tempo values distributed across all tracks? (Are songs generally fast, slow, or is there a wide spread?)
- What is the distribution of acousticness scores? (Does the dataset lean towards acoustic or electronic sounds?)
- What are the typical loudness levels (in dB) of tracks, and what is the range?
- How is danceability distributed? (Are most songs highly danceable, or is there a mix?)
- What is the distribution of energy levels in the dataset? (Are songs generally high or low energy?)
- What are the most common time_signatures found in the music?
- What is the distribution of speechiness? (Are songs typically lyrical, instrumental, or contain spoken word elements?)
- What is the overall distribution of valence scores? (Are most songs positive/happy, or is there a wide spread of moods?)
- What is the distribution of instrumentalness? (Does the dataset lean toward tracks with vocals or without?)
- How are liveness scores distributed? (Are most songs recorded in a studio or in a live setting?)
- What are the most common values for mode (major or minor key)?
- What are the median and quartile values for popularity and duration_ms? (This gives a clearer picture of the typical song than just the average, which can be skewed by outliers.)
- What is the modal (most frequent) energy level in the dataset?

- What is the distribution of songs across different language categories?

2. Bivariate Analysis

- Is there a correlation between a song's duration_ms and its popularity? (Are shorter or longer songs more popular?)
- How does danceability relate to popularity? (Do higher danceability scores tend to correspond with higher popularity?)
- What is the relationship between energy and popularity? (Are high energy tracks generally more popular than low-energy ones?)
- Does loudness have a noticeable impact on popularity? (Are louder mixes preferred by listeners?)
- Is there a relationship between acousticness and popularity? (Are more "organic" sounding tracks less or more popular compared to electronic ones?)
- **Valence vs. Popularity:** How does a song's valence (its musical positivity/mood) relate to its popularity? Do more cheerful or more somber tracks tend to be more popular?
- **Instrumentalness vs. Popularity:** Is there a relationship between a track's instrumentalness and its popularity? (Does a lack of vocals impact a song's popularity?)
- **Liveness vs. Popularity:** How does liveness relate to popularity? (Are songs recorded in a live setting generally more or less popular than studio recordings?)
- **Tempo vs. Popularity:** What is the relationship between a song's tempo and its popularity? (Are faster or slower songs typically more popular?)
- **Language vs. Popularity:** Does the song's language influence its popularity? (Are songs in certain languages consistently more popular than others?)
- **Key and Mode vs. Popularity:** How does popularity differ across various musical keys (key) and modes (mode)?
- **Time Signature vs. Popularity:** Do specific time_signature values correspond to a higher or lower average popularity?

3. Multivariate Analysis

- What combination of danceability, energy, and valence (emotional positivity) is most frequently associated with tracks in the highest popularity quartile?
- Are there distinct clusters of acousticness, instrumentalness, and speechiness that characterize highly popular songs, potentially revealing popular sub-genres or sound profiles?
- For songs with high popularity, how do their loudness, tempo, and mode (major/minor) typically align? (Can we identify a "popular mix recipe"?)
- How do the average values of danceability, energy, and valence for popular songs differ across various language categories? (This can help identify if the "recipe" for a popular song is culturally or linguistically specific.)
- Are there distinct clusters of tracks based on a combination of their acousticness, instrumentalness, and speechiness that correlate with higher popularity? (This could reveal popular sub-genres or sound profiles.)
- For songs in the highest popularity quartile, how do their loudness, tempo, and mode (major/minor) typically align? (This can help identify a "popular mix recipe.")
- What is the relationship between a song's popularity and a combination of its key, mode, and time_signature? (Does a song in a specific key and mode, with a particular time signature, have a higher chance of being popular?)
- How do the duration_ms and liveness of songs with high popularity change across different year decades? (This can help identify long-term trends in song length and recording style for popular music.)

4. Timeseries Analysis

- How has the average popularity of songs evolved over years? (Are songs becoming generally more or less popular over time?)
- Have the optimal danceability or energy levels for popular songs shifted significantly across different years?
- Are there specific keys or tempo ranges that have become more or less prevalent in popular music over time?

- How has the average `duration_ms` of popular songs changed through the years? (Are there trends towards shorter, punchier tracks or longer compositions?)
- Are there observable trends in acousticness or instrumentality in popular music across different years, indicating a shift in production styles?
- How has the average **valence** (musical positivity) of songs evolved over the years? (Are tracks generally becoming more cheerful or somber?)
- How has the average **loudness** of songs changed over time? (Has the "loudness war" had a quantifiable effect on music production?)
- Are there observable shifts in the average **liveness** of popular music? (Is there a trend towards more live-sounding or studio-perfect tracks?)
- What are the trends in the prevalence of different **language** categories over the years? (Are songs in certain languages becoming more or less common?)
- How has the average **speechiness** of songs changed? (Is there a trend towards more lyrical, rap-heavy, or instrumental tracks?)
- How has the relationship between two features, such as **danceability** and **energy**, evolved over time? (Does the "formula" for a high-energy dance track change with the years?)

5. Examples of Hypothesis generations based on Descriptive Statistics

- **Distribution Skew:** The median popularity score is significantly lower than the mean popularity score, suggesting the distribution is heavily skewed by a small number of very popular songs.
- **Central Tendency:** The majority of tracks have a `duration_ms` between 3 and 4 minutes, as indicated by the interquartile range (IQR) and a histogram visualization.
- **Correlation:** There is a strong positive correlation (e.g., a Pearson correlation coefficient greater than 0.7) between a song's danceability and its energy scores.
- **Categorical Grouping:** The average popularity of songs in English is descriptively higher than the average popularity of songs in Spanish within this dataset.

- **Outlier Presence:** The box plot for loudness reveals several data points that are statistically considered outliers, with decibel levels significantly lower or higher than the typical range.
- **Modal Analysis:** The most frequently occurring time_signature in the dataset is 4/4, as shown by the mode and a count plot.

6. How to present recommendations

Sample Recommendations for a Music Director/Mixing Engineer

- **Focus on an "Energetic & Danceable" Sound Profile:**
 - **Insight:** Our analysis shows that songs within the top 25% of popularity tend to have a consistently high **danceability** (e.g., above 0.7) and **energy** (e.g., above 0.8). These tracks strike a balance between a strong beat and an intense sound.
 - **Recommendation:** Prioritize producing tracks that fall within this defined "energetic & danceable" quadrant. For new song concepts, aim to maximize both of these features to align with the characteristics of popular music.
- **Master for a Modern, Impactful Mix:**
 - **Insight:** The time series analysis revealed that the average **loudness** of popular tracks has steadily increased over the decades, while the average **duration_ms** has slightly decreased. Modern popular music is louder and more concise.
 - **Recommendation:** When mixing and mastering new tracks, target a loudness level that is competitive with current chart-toppers. Additionally, experiment with a shorter, more efficient song structure to increase listener engagement and streaming completion rates.
- **Leverage Shifting Language Trends:**
 - **Insight:** An analysis of **language** by **year** shows that while English remains dominant, songs in other languages (such as Spanish and Korean) have seen a significant, rapid increase in popularity over the last 10 years.
 - **Recommendation:** Explore collaborations with artists who create music in these rising languages or consider producing multilingual versions of key tracks to tap into rapidly growing international markets.

- **Target a "Lyrical" vs. "Instrumental" Niche:**
 - **Insight:** The distribution of **speechiness** is heavily bimodal, with most songs either having very low or very high speechiness. This suggests a clear divide between purely instrumental/vocal tracks and lyrical/rap-heavy tracks.
 - **Recommendation:** Deliberately decide which niche a new track will target. For instrumental or "non-speechy" tracks, focus on maximizing other features like energy and valence to drive popularity, as that's where success lies for this segment.